Anna-Drishti — Science Brief

Everything you need to say, and every question you will be asked.

Krittika Parida & Netra R · for today's online review

READ THIS BOX FIRST

You are not being asked to explain the software. You are being asked **two things only**: is the biology real, and is the comparison fair. Krittika owns the first. Netra owns the second. Nothing else on this call is your responsibility.

Everything below is written in plain language on purpose. You do not need to sound technical. You need to sound like you know what you are talking about — and the way to do that is to explain it simply.

THREE RULES FOR TODAY

1. Short answers win. Answer in two or three sentences and stop. If they want more they will ask. Long answers sound nervous and invite attack on details nobody asked about.

2. Never bluff a number. If you do not know, say so and offer what you do know. Judges respect that and punish guessing. There is a script for this on the last page.

3. Volunteer your limits before they find them. Saying "here is what we cannot do" makes everything else you say more believable. This is the single most powerful thing either of you can do today.

WHERE THE JARS ACTUALLY STAND TODAY

The jars were sealed yesterday evening. That is roughly **fifteen hours**, not the two days originally planned. This matters, so know exactly what you can and cannot claim:

CLAIM	CAN WE SAY IT?	WHY
"We measured water activity in two jars"	**YES**	The air above grain reaches balance with the grain in a few hours. This is ready now.
"One jar is above the growth threshold, one below"	**YES**	That is exactly what the reading shows.
"We watched grain spoil"	**NO**	Fungus needs days. Nothing has grown yet.
"We measured CO_2"	**NO**	We could not get a CO_2 sensor in time. We have none.

> **IF ANYONE ASKS WHETHER THE JARS HAVE MOULDED**
>
> “Not yet — we sealed them yesterday evening. At this water activity our model predicts growth takes several days, so seeing nothing after fifteen hours is exactly what we expect. What we have verified today is the *input* to the prediction, not the outcome.”

That answer is strong because it turns a gap into evidence that the model is behaving consistently. Use it confidently rather than apologetically.

Krittika — the biology

Your job: convince them the biological numbers are real, and that we know where every one of them came from.

PART 1 · THE ONE IDEA YOU MUST BE ABLE TO EXPLAIN

Water activity — and why moisture content is not enough

Not all the water inside a grain kernel is usable. Some of it is chemically stuck to the starch and protein. Fungi cannot touch that part. Only the **free** water — the water that can move and evaporate — is available for a fungus to grow on.

Water activity is simply the measure of how much of the water is free. It runs from 0 to 1. Pure water is 1.0. Bone-dry material is 0.

THE ANALOGY TO USE IF THEY LOOK UNCONVINCED

Two people each have ₹1,000. One has it as cash in hand. The other has it locked in a fixed deposit. Both have the same *amount* of money, but only one of them can spend it today. Moisture content is the total. Water activity is the spendable part.

This is also why honey and jam do not spoil despite being full of water — the sugar locks it all up.

SAY IT LIKE THIS

> "Fungi do not respond to how much water is in the grain. They respond to how much of that water is *available*. That is water activity, and it is what actually decides whether mould grows. Two batches at the same moisture percentage can behave completely differently at different temperatures, because temperature changes how much of that water is free."

This is why our system converts humidity into water activity instead of reporting a moisture percentage. It is the single most important scientific decision in the project, and it is yours.

PART 2 · THE NUMBERS YOU OWN

NUMBER	WHAT IT MEANS, PLAINLY
0.82	**The floor.** Below this water activity, *Aspergillus flavus* physically cannot pull enough water out of the grain to grow at all. Not "grows slowly" — cannot grow. This is the line our whole system defends.
0.85	**The toxin line.** The fungus can grow from 0.82, but it does not start making aflatoxin until conditions are wetter still. Toxin production needs a narrower window than growth does.
25–35 °C	**The toxin temperature band**, with 25–30 being the worst. Exactly the temperature of an Indian godown.

NUMBER	WHAT IT MEANS, PLAINLY
`12–43 °C`	**The growth temperature range**, fastest around 33. Below 12 or above 43 it stops.
`17 days`	What our model predicts for mould onset at water activity 0.87 and 30 °C. This is the kind of number a judge will pick on, so know it.

PART 3 · WHY AFLATOXIN, AND WHY PREVENTION IS THE ONLY OPTION

Aflatoxin is a poison made by the fungus. It causes liver damage and is a known carcinogen. The critical fact is this:

> SAY IT LIKE THIS — THIS IS YOUR STRONGEST LINE
>
> “You can dry grain and stop the fungus growing. You cannot remove the toxin it already made. Drying does not destroy it, and normal cooking does not destroy it — it is heat stable. Once aflatoxin is in a consignment, that grain’s fate is decided.”
>
> “That is why this problem has to be predictive. The only useful intervention happens *before* the fungus grows. A system that tells you the grain is already contaminated has told you nothing you can act on.”

PART 4 · WHY CO_2 APPEARS BEFORE HEAT — THE BIOLOGY HALF

Mohit owns the physics of this. You own the biology of *why there is a signal at all*.

Grain is alive. So are the fungi on it. Both respire — they consume sugars and oxygen, and release carbon dioxide, water and heat. When a wet pocket forms, respiration speeds up sharply, and that single process produces **both** the CO_2 and the warmth.

> THE ANALOGY
>
> A compost heap. It gets warm in the middle and it gives off gas, because things are metabolising inside it. A spoiling pocket in a grain stack is the same event, much smaller and much slower.

> SAY IT LIKE THIS
>
> “Respiration produces gas and heat from the same reaction. The gas escapes through the air gaps between kernels and reaches a probe within about a day. The heat has to conduct through the kernels themselves, and grain is a good insulator — that takes about fifteen days over the same distance. So the same biological event announces itself as gas long before it announces itself as warmth.”

PART 5 · EVERY QUESTION YOU WILL GET, AND WHAT TO SAY

Q How do you know this reflects real spoilage?

Three sources, and one of them is our own grain.

“Three things. The growth model uses published cardinal values for *Aspergillus flavus* — the minimum, optimum and maximum conditions are from the predictive mycology literature, not invented. The equation converting humidity to water activity is a published standard equation with published coefficients for maize. And we measured our own jars this morning, and the readings agree with what the model expects.”

Q Have you actually grown the fungus? Is any of this validated in a lab?

Not yet, and we say so. Here is the plan.

“Not yet. What we have validated today is the measurement side — we can read water activity in real grain and interpret it correctly. The biological validation is a four-experiment programme: a moisture ladder with fungal plating at intervals, insect inoculation with manual counts, a concealed hotspot trial, and intervention verification. I have laboratory access and my mentor’s approval for it.”

Do not apologise for this. Every honest team is at this stage. Teams that claim lab validation they do not have get caught.

Q Where did 0.82 come from?

Published mycology. It is a well-established threshold.

“It is the reported minimum water activity for *Aspergillus flavus* growth in the storage-fungi literature. The reported range is 0.82 to 0.85 and we deliberately took the lower, more cautious end, because a monitoring system should alarm early rather than late.”

Q Why not just measure moisture content? Farmers already have moisture meters.

Because moisture content alone does not tell you if mould will grow.

“A handheld meter gives you one average number for a whole bag. Two problems. First, spoilage is local — the average is fine while a pocket inside is dangerous. Second, the same moisture content is safe at 20 degrees and unsafe at 35, because temperature changes how much of that water is available to the fungus. Water activity accounts for both.”

Q Your equation is for maize. What if the godown stores wheat or rice?

Different coefficients, same equation. It is a configuration change.

“The relationship between humidity, temperature and moisture is described by the same standard equation for all cereals — only the coefficients change, and they are published for each commodity. We fixed our scope to one commodity for validation deliberately. Adding a second is changing three numbers, not redesigning anything.”

Q Can your system detect aflatoxin itself?

No. Say this clearly and without hedging.

"No, and we are explicit about that everywhere in our system. We measure and predict the *conditions* under which the fungus grows and produces toxin. Confirming the toxin itself needs a laboratory assay. A system claiming to read toxin levels off a humidity sensor would be scientifically indefensible, and we would rather say that plainly than imply otherwise."

Q Is 12 % moisture realistic for stored grain in India?

Yes — it is the acceptance specification.

"Twelve percent is around the moisture level grain is accepted into public storage at, precisely because it is safely below the threshold where fungi can grow. Our simulation starts from sound, correctly dried grain and then introduces a localised problem — which is how spoilage actually starts. It does not start with bad grain."

Q Why should I believe "mould in 17 days"?

It comes from a standard model, and it is a prediction we can be checked on.

"It comes from a cardinal-value growth model, which is the standard approach in predictive mycology — you take the published minimum, optimum and maximum for water activity and temperature, and it gives you a growth rate. We are not claiming precision to the day. We are claiming the right order of magnitude, and that the trend responds correctly to conditions. And it is falsifiable — our lab programme is designed to test exactly this."

KRITTIKA — THINGS NOT TO SAY

"We detect aflatoxin" — we do not.
"We validated this in the lab" — not yet.
"The grain spoiled in our jar" — it has not, it has been fifteen hours.
"This is accurate to the day" — claim the trend, not the precision.

Netra — the comparison, and why it is fair

Your job: defend the claim "conventional inspection found it on day 42, we found it on day 4." This is the most likely thing to be attacked today.

WHY YOUR PART MATTERS MOST

Our headline number is a comparison. A comparison is only as strong as the thing it compares against. If our model of ordinary inspection looks rigged, the whole claim collapses — and a rigged comparison does more damage than no comparison at all.

The good news: **our baseline is genuinely defensible, and in places we deliberately made it generous.** Your job is to show that, not to hide it.

PART 1 · HOW GRAIN IS ACTUALLY INSPECTED TODAY

An inspector walks around a stack of bags. They look at the outside for staining, mould, insect holes and damage. Then they push a **spear** — a hollow metal tube with slots along it — into some bags, pull out a sample, and examine it for insects, mould and smell.

This is careful, skilled work. It is also physically limited in three ways:

1. **It only reaches the outside.** A spear reaches roughly 40 centimetres into a stack. A godown stack is metres deep. The middle is simply not reachable.
2. **Problems are not spread evenly.** Insects and moisture gather in pockets — near dust, near warmth, near a leak. A clean sample from one spot tells you nothing about a pocket two metres away.
3. **You cannot see the early stages.** Insect eggs and larvae develop *inside* the kernels. A sample with no adult insects in it can still be heavily infested.

THE ANALOGY TO USE

It is like checking whether a building has a problem by only ever looking at the outer wall. You will find the problems that have reached the surface. You will not find the one starting in the middle of the third floor — not because you were careless, but because you could not reach it.

PART 2 · THE NUMBERS YOU OWN

WHAT	OUR VALUE	HOW TO DEFEND IT
Inspection interval	`every 14 days`	Fortnightly is at the **generous** end of real practice. Many facilities are monthly or slower. We did not pick a slow interval to flatter ourselves.
Spear reach	`0.4 m`	The physical limit of a bag spear into a stacked wall of bags. **This is your key number.**

WHAT	OUR VALUE	HOW TO DEFEND IT
Chance of finding a pocket that is in reach	`60 %`	Because problems cluster, sampling a few bags may miss a nearby pocket. Six times out of ten is a reasonable, not harsh, assumption.
Surface mould	`always found`	We assume inspectors **never** miss visible mould on an outer bag. That is us being generous to the baseline.
Musty smell / warm bags	`also found`	We **added** a second way for ordinary inspection to succeed. See below — this is your best defensive point.

PART 3 · YOUR SINGLE STRONGEST ANSWER

When they ask whether the baseline is unfair, **do not defend the inspection interval.** Go straight to physics.

> LEARN THIS ONE PROPERLY — IT IS THE ANSWER THAT WINS THE ARGUMENT
>
> "We are not modelling inspectors as careless. We are modelling a physical limit. A spear reaches about forty centimetres. The hotspot in our demonstration sits one point six metres inside the stack. No amount of diligence gets a spear to that location."
>
> "That is not a criticism of inspection. It is precisely the reason an interior probe needs to exist."

This reframes the whole thing. You are no longer arguing about whether inspectors are good at their job — you are pointing at a wall they cannot see through. That is a much stronger position, and it is true.

PART 4 · THE POINT THAT PROVES WE WERE NOT RIGGING IT

Originally, our model had conventional inspection **never** finding the deep pocket at all — because a spear cannot reach it. Technically correct, but it looked unfair.

So we **added** a second route for ordinary inspection to succeed: once a colony is well established, the stack smells musty and the bags feel warm, and we assume the next routine inspection catches that. That is what produces day 42 instead of "never."

> SAY IT LIKE THIS IF THEY PUSH ON FAIRNESS
>
> "We actually went out of our way to make the comparison *less* favourable to ourselves. Our first model had ordinary inspection never finding this pocket at all, which is arguably what the guidance implies. We thought that looked unfair, so we added a second way for inspection to succeed — smell and warmth once the problem is established. That is where day 42 comes from. Without it, the gap would look much bigger."

Almost no team volunteers that they weakened their own headline number. It is very hard to attack someone who has already done the attacking for you.

Q Isn't your baseline unfairly pessimistic?

Use Part 3, then Part 4. Reach first, generosity second.

Lead with the 0.4 metres versus 1.6 metres point. Then mention that we added a detection route to help the baseline. Stop there.

Q A good inspector would notice something. Are you not underestimating them?

We assume they never miss anything visible. The problem is not visible.

"In our model, an inspector never misses mould on an outer bag — not once. We are not assuming poor skill. We are pointing out that a pocket in the middle of a stack has not reached any surface yet. There is nothing there for a skilled person to notice."

Q Why not just inspect more often?

More frequent inspection of the outside does not reach the inside.

"Inspecting twice as often does not extend a spear's reach. You would double the labour and still not reach the middle. Frequency does not fix a coverage problem — and probing more also damages more bags."

Q What about temperature cables? Those already exist.

They are for silos, and heat arrives late.

"Two reasons they do not solve this. They are designed for bulk silos, not the bagged stacks that dominate Indian godowns — different geometry, different access. And heat conducts through grain very slowly, so the signal arrives late and weak. That is exactly why we lead with gas instead of temperature."

Q Have you visited a real godown? Have you seen this happen?

Answer honestly. Do not invent a site visit.

If you have not: "Not yet — our model of inspection practice is built from published guidance on storage and inspection of bagged grain rather than from our own site observation. A facility visit is in our plan, and it is exactly the thing that would let us replace assumptions with measured intervals."

Naming the gap and naming the fix is a strong answer. Inventing a visit is fatal if followed up.

Q Insects or mould — which are you actually solving?

Both, through one signal, but mould is what we demonstrate today.

"Both show up as increased respiration, so the same gas measurement sees either. What we demonstrate today is the mould and mycotoxin pathway, because that is the irreversible one. Insect detection through sound is in our design and is the next thing we build."

Q Would FCI or a cooperative godown actually adopt this?

The cost and the offline design are the answer.

"Under twenty thousand rupees for a prototype, and it runs completely offline because rural godowns cannot be assumed to have connectivity. Existing commercial systems are built and priced for large Western silos. The adoption route is one facility, documenting lead time and chemical use against their current practice, then extending outward."

NETRA — THINGS NOT TO SAY

"Inspectors do a bad job" — never. It is a physical limit, not a criticism.
"We visited a godown" — only if true.
"Inspection never works" — it works for surface problems, which is our point.
Do not get drawn into arguing about the 14-day interval. Move to the 0.4 metre reach.

Both of you — the last page

Read this in the five minutes before the call.

WHEN YOU DO NOT KNOW THE ANSWER

> USE THIS EXACT STRUCTURE
>
> "I do not have that figure in front of me, so I would rather not guess. What I can tell you is …" — then give the closest thing you do know.

This costs you nothing. Guessing wrong and being corrected costs you the whole panel's confidence, in every answer that follows.

ONLINE CALL DISCIPLINE

- **Stay muted until your name is said.** Two people answering at once sounds like a team that has not prepared.
- When you are asked something, **unmute and pause for one second** before speaking. Audio cuts off the first word otherwise, and your first word is usually the important one.
- Keep this document open on screen, not printed in your lap. Looking down repeatedly reads as uncertainty on camera; glancing at a second window does not.
- If a question is for someone else, **do not add to it.** Chiranjib will bring you in.

THE THREE SENTENCES THAT MATTER MOST TODAY

Krittika: "You can dry grain and stop the fungus, but you cannot remove the toxin it already made. That is why this has to be predictive."

Netra: "A spear reaches forty centimetres. Our hotspot is at one point six metres. That is not a criticism of inspection — it is the reason an interior probe needs to exist."

Either of you: "We do not measure the toxin. We predict the conditions that create it, and we say so in every output."

> LAST THING
>
> You two are the reason this project is credible. Three software students could have built the dashboard. They could not have told a judge where 0.82 comes from, or why a spear cannot reach the middle of a stack. That is the part nobody else in the room can replicate — so answer as the person who owns it, not as someone reciting it.